AmbuGo v8: How much does ambulance signal priority cost everyone else?

A verified simulation study of emergency-vehicle preemption in a mixed-traffic signalised district

October 2026 — synthetic study; no field data

Abstract

We ask two linked questions. How much can signal priority shorten an ambulance's trip through a congested signalised district? What does that priority cost ordinary traffic, and how does the cost depend on network load? We built a 1-second mesoscopic spatial-queue simulator of a 16-junction synthetic district with spillback, mixed vehicle classes in passenger-car units (PCU), Webster-optimised fixed-time signals, noisy sensors and a stochastic demand process. We verified it against Webster's (1958) delay formula (errors of −3.5% to +6.6%) and against an independent fluid queueing model. We then compared one no-priority baseline with five priority policies in a paired, common-random-number experiment: 14,400 held-out simulations across six load levels (critical degree of saturation X = 0.60–0.95) and two driver-yielding regimes. Policy parameters were tuned on separate training replications.

Four findings stand out:

- **Phase skipping is the lever that matters.** Preemption with phase skipping cuts mean ambulance travel time by 125–164 s (35–43%). Classic extension/truncation priority, the v7 design, achieves only 27–98 s, and skipping costs no more network delay than it does.
- **The cost to ordinary traffic follows a 1/(1−X) law.** It is statistically zero for X ≤ 0.7. It rises to about 1 vehicle-hour per mission at X = 0.9 and 1.5 at X = 0.95. The exchange rate climbs from about 10 to 32 vehicle-seconds of ordinary delay per ambulance-second saved. Above X ≈ 0.9 the disturbance is not absorbed within 33 minutes.
- **The refinements we expected to help did not.** Queue-aware triggering and post-priority compensation produced no measurable improvement. Queue-aware re-routing helped only near saturation (−9 s at X = 0.9). Even perfect state information is worth at most about 6 s, so the expensive particle-filter estimator planned in v7 is not justified.
- **The protections can't always hold.** The 180 s wait caps are infeasible for the baseline itself above X ≈ 0.8, so protections must be framed as soft targets or load-dependent.

We also show that a 1-second signal nudge at a single junction moves network delay by as much as the ambulance does. Network-level effects of a single intervention are therefore only detectable with paired designs of order 10^2 replications.

1. Research question and why it is hard

Emergency-vehicle preemption (EVP) gives an ambulance green lights along its route. The benefit to the ambulance is easy to see. The cost is diffuse: conflicting approaches lose green, queues grow, and the backlog spreads through the network after the ambulance has gone. Three things make this cost hard to measure:

1. **It depends on load, nonlinearly.** The delay from losing Δ seconds of green on an approach with saturation flow s scales roughly as $(s\Delta)^2/(c(1-X))$. The same intervention is free at X = 0.6 and expensive at X = 0.95 (§6.2, and test T5 in §4).
2. **It is small relative to the noise.** Traffic networks are sensitive to small perturbations: re-ordering a few vehicles changes every later interaction. Comparing a single priority run with a single no-priority run therefore measures mostly noise (§6.7).
3. **The baseline matters.** A badly timed baseline creates its own congestion, and the measured cost of priority then mostly reflects that congestion rather than the priority (§2).

2. From v7 to v8: what was wrong and what changed

v7 was an input fixture with no simulator, optimiser or results. A referee review identified the problems below. Every item is addressed in v8 and backed by evidence in this report.

#	v7 issue	v8 change	Evidence
1	Random greens (20/24/28 s) made 8/51 approaches oversaturated at mean demand	Demand calibrated to a declared target X; Webster cycle and demand-proportional splits per junction	Fig. 5: at identical demand the v7 plan accumulates about 275 veh/h and strands 283 vehicles at the boundaries after 2 h; Webster is stationary
2	Results were an unacknowledged function of load	Load X is the primary experimental axis (6 levels)	Figs. 1–4
3	2 s “near-fastest” tolerance far below Monte Carlo noise (SE ≈ 9 s)	Paired common random numbers, 200 replications per cell, bootstrap CIs, sign-flip tests, Holm correction	`tables/main_results.csv`
4	32 scenarios cannot certify a 5% failure rate	200 replications; exact Clopper–Pearson intervals	`tables/tail_risk.csv`
5	Optimiser’s curse (best of 528 on the same 32 scenarios)	Tuning on training replications 5000–5039; evaluation on held-out replications 0–199	`tuned_parameters.json`
6	Action set capped benefit (no phase skip; ≥ 56–69 s forced wait per junction)	Preemption with phase skip (P2), protected by a pedestrian/vehicle wait guard	P2 − P1 = −65 to −103 s (§6.1)
7	Queue ahead of the ambulance can exceed one green	Queue-aware trigger (P3) tested explicitly	No measurable gain (§6.5): a negative result

#	v7 issue	v8 change	Evidence
8	No driver yielding	Class-weighted yielding model, $\rho \in \{0, 0.5\}$	§6.8
9	Compute target infeasible ($\approx 10^{10}$–10^{11} ops/s)	Rule-based controllers; value-of-information bound	13 µs median decision (§6.10); perfect information worth ≤ 6 s (§6.6)
10	`routes.csv` leaked hidden initial state ($r = 0.85$)	Strict `public/` vs `hidden/` export	Test T6: public files byte-identical across replications
11	One RNG stream (no common random numbers across design variants)	SHA-256 namespaced streams	Test T1b
12	All-or-nothing routing left 12/51 approaches empty	Up to 4 near-shortest routes per OD pair with a logit split	0 empty approaches
13	600 s warm-up	MSER-5 warm-up = 1,405 s ($X = 0.8$) and 1,625 s ($X = 0.95$); dispatch after 1,800 s	§5
14	Static packed initial snapshot	Initial state reached by simulated warm-up; exported as `hidden/state_at_dispatch.csv`	Fixture
15	788 of 803 checks were tautological	Runtime invariants plus external validation	§4
16	Buses counted like motorcycles	PCU-based storage and discharge	`network.py`
17	AR(1) demand shock not stationary at the start	Stationary initialisation	`scenario.py`
18	“Correlated” outages drawn independently	Spatially clustered common-cause outage	`sim.py`
19	Protection definitions ambiguous	Per-vehicle stop-line wait ≤ 180 s and pedestrian wait ≤ 180 s, enforced by an earliest-deadline guard shared by **all** policies	§6.4

Two corrections to the v7 review are also on record:

- **The review’s cost estimate was too high.** Its closed-form lost-green cost overstates the exact fluid-model cost by 1.3–1.6×, although the $1/(1-X)$ scaling holds (Fig. 6, right).
- **The v7 demand level was not the problem.** Under Webster timing it gives $X \approx 0.79$. The oversaturation came from the signal plan.

3. Model

Network. The v7 geometry is kept unchanged: 16 signalised junctions, 8 boundary nodes, 59 directed links of 260–472 m, with one- and two-lane links. Saturation flows are 0.40–0.48 PCU/s/lane and jam density is 0.16 PCU/m/lane. Free-flow speeds are 30 and 40 km/h.

Demand. Each origin releases vehicles as a Poisson process whose rate is modulated by a common AR(1) log-normal shock ($\rho = 0.75$ per minute, $\sigma = 0.18$, stationary). Destinations follow a gravity model with hospital attraction. Each OD pair has up to 4 routes within 1.25× of the free-flow time, split by a logit model ($\theta = 0.05$ s^{-1}). The class mix is 53% cars, 30% motorcycles, 10% autorickshaws, 4% buses and 3% goods vehicles, with PCU values 1.0 / 0.5 / 1.0 / 3.0 / 2.5. A single scale factor κ is solved by bisection so that the critical approach reaches the target X under its Webster plan.

Signals. Each green serves one protected approach (3 s amber, 2 s all-red), followed by an exclusive pedestrian stage (7 s walk, 16 s clearance, 2 s all-red). Webster cycles are clipped to 60–150 s, with greens of 8–60 s. The **base controller**, used by every policy including the baseline, adds an earliest-deadline-first guard. At each stage choice it projects the coming stage sequence. If any waiting vehicle (detected by a stop-line presence detector) or the pedestrian stage would exceed 180 s, it serves the most urgent stage first.

Simulator. Each link is a FIFO queue with finite PCU storage, so spillback and gridlock emerge from the dynamics. Vehicles reach the stop line after the free-flow time and discharge at the saturation flow during green (after a 2 s start-up loss) and amber. Multi-lane approaches expose one head per lane. Boundary demand waits in external queues, and that waiting counts as delay. Incidents (probability 0.5 per replication) reduce one lane of a two-lane link to 45–65% capacity for 4–8 minutes near the dispatch time. Sensors are 5 s link-count cameras with $\{-2\ldots+2\}$ noise, Markov dropout and a clustered outage.

Ambulance. The ambulance obeys the signal, has no speed privilege, and uses no opposing lanes. It is dispatched to the hospital (BE1) from one of six boundary origins at a random time 1,800–2,100 s into the run. With yielding parameter ρ, each queued vehicle ahead of it pulls aside with probability $\rho \cdot w$, where w is 1.0 for motorcycles, 0.8 for autorickshaws, 0.6 for cars and 0.2 for buses and goods vehicles.

Policies.

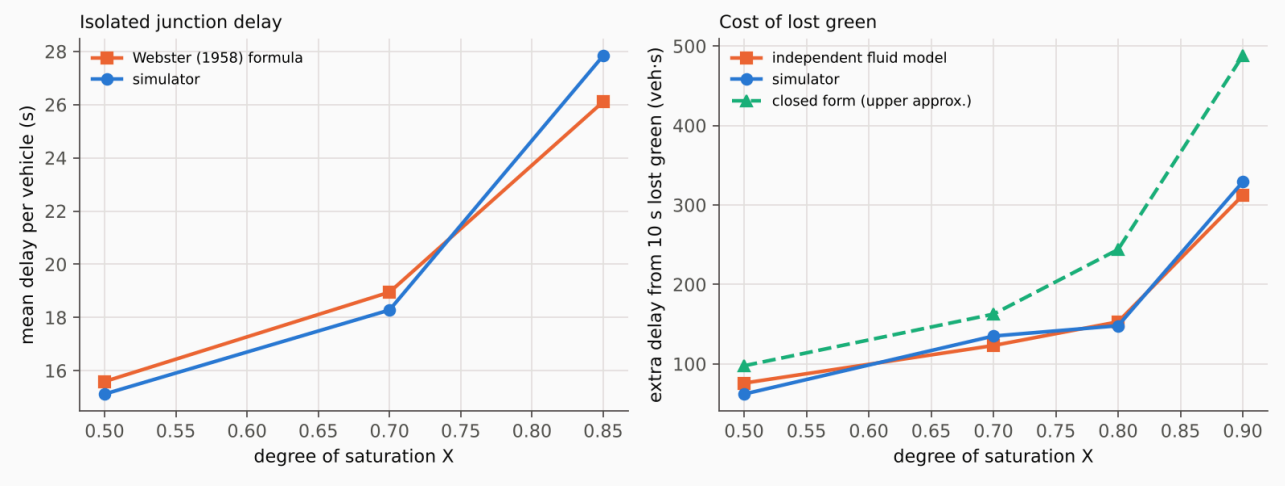
Policy	Logic
P0	No priority (base controller only)
P1	Classic EVP, the v7 design: queue-blind trigger when the ambulance's free-flow ETA ≤ H; extends the ambulance green by up to 10 s; truncates conflicting greens to the 8 s minimum; no skipping
P2	Preemption: P1 plus a jump to the ambulance approach after the current clearance (skipping the walk stage only if its wait clock allows); holds the green until the ambulance crosses (cap 60 s)
P3	Queue-aware preemption: as P2, but the trigger subtracts the estimated time to discharge the queue ahead of the ambulance and adds upstream queue delays, plus a margin; hold cap 90 s
P3C	P3 plus service-deficit compensation: approaches that lost green get it back, limited by the guard's slack
P4	P3C plus queue-aware ambulance re-routing at every link entry (switches route only for a gain > 10 s)

4. Verification

All checks are in `tests/run_tests.py` and the results in `results/verification.json`. **8/8 pass.**

Test	What it establishes	Result
T1a–c	Bit-identical reruns; changing incident draws leaves the demand realisation untouched; replications differ	pass
T2	Runtime invariants on 24 stress runs (43,200 audited ticks, X up to 0.95, forced incidents, yielding): exact vehicle conservation every tick, PCU bookkeeping, storage never exceeded, ambulance never crosses on red, minimum greens, no green > 90 s	0 violations
T3	Isolated junction with Poisson arrivals against Webster's delay formula	−3.0%, −3.5%, +6.6% at X = 0.5, 0.7, 0.85
T4	Extra delay from 10 s of lost green against an independent fluid cumulative-curve model	−18%, +10%, −3%, +5% at X = 0.5–0.9
T5	Cost × (1−X) roughly constant (the 1/(1−X) law)	spread 21%
T6	Public fixture files independent of plant truth (no information leakage)	identical hashes across replications

During development these tests caught two real bugs. The ambulance's arrival was being counted as an ordinary vehicle completion, biasing delay by about 0.5 vehicle-hours. And an early version applied the wait guard only inside priority policies, which handed them a hidden advantage of adaptive control. Both are fixed. We report them because they were exactly the size of the effects being measured.



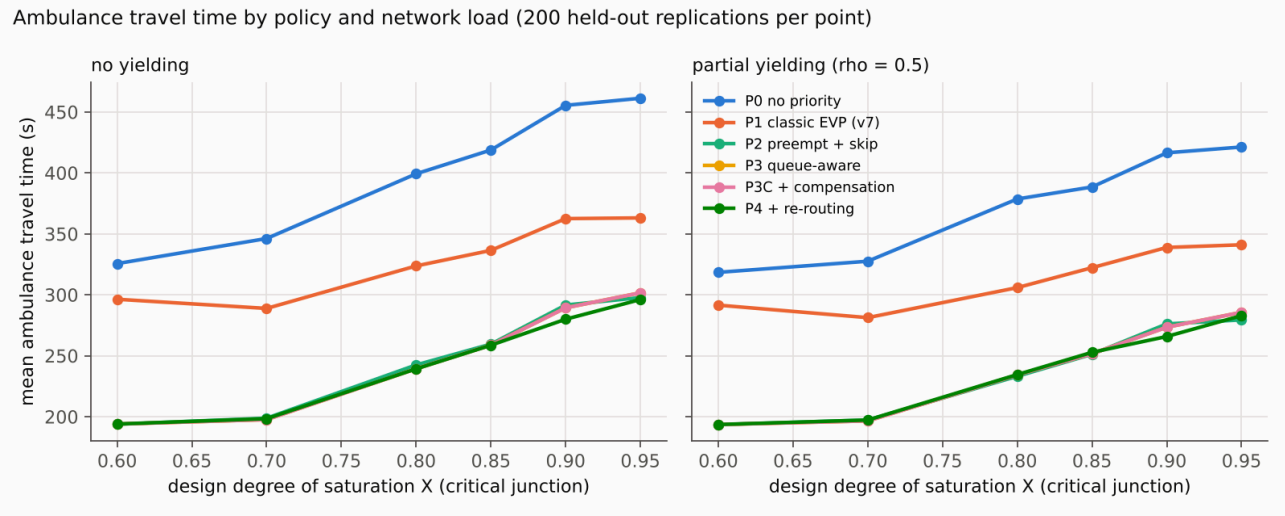
Verification against Webster's formula (left) and an independent fluid model (right).

5. Experimental design (fixed before evaluation)

- **Factors:** load X ∈ {0.60, 0.70, 0.80, 0.85, 0.90, 0.95} × yielding ρ ∈ {0, 0.5} × policy {P0, …, P4}.
- **Replications:** 200 held-out replications (seeds 0–199) per cell, 14,400 simulations in total. Every policy faces the identical demand, incident, sensor and ambulance realisation (common random numbers), so all comparisons are paired.
- **Warm-up and horizon:** simulations start empty. Dispatch happens after the MSER-5 warm-up (1,800–2,100 s), and the horizon is 4,200 s, leaving ≥ 2,100 s after dispatch.
- **Tuning:** on training replications 5000–5039 at X ∈ {0.8, 0.9} × both ρ values. The rule was: minimise mean ambulance time, and among candidates within one paired standard error of the best, take the one with the smallest ordinary-delay increase. It selected H = 30 s for P1 and P2 and margin = 10 s for P3/P3C/P4. Longer look-aheads were worse on both criteria: H = 60 gave +48 s ambulance time and +1.8 vehicle-hours for P2.
- **Metrics:**
  - ambulance travel time: mean, median, P90, $CVaR_{90}$, and P(> 300 s) with exact CIs;
  - change in ordinary delay = ∫(N_policy − N_P0) dt from dispatch to horizon. This equals the change in total vehicle delay by Little's law under common random numbers. Also reported in person-hours using occupancies;
  - wait-cap breaches;
  - ensemble recovery.
- **Inference:** paired percentile bootstrap (10,000 resamples), two-sided sign-flip randomisation tests, Holm correction over the 60 policy-by-cell delay tests.

6. Results

6.1 Phase skipping, not trigger sophistication, drives the ambulance benefit



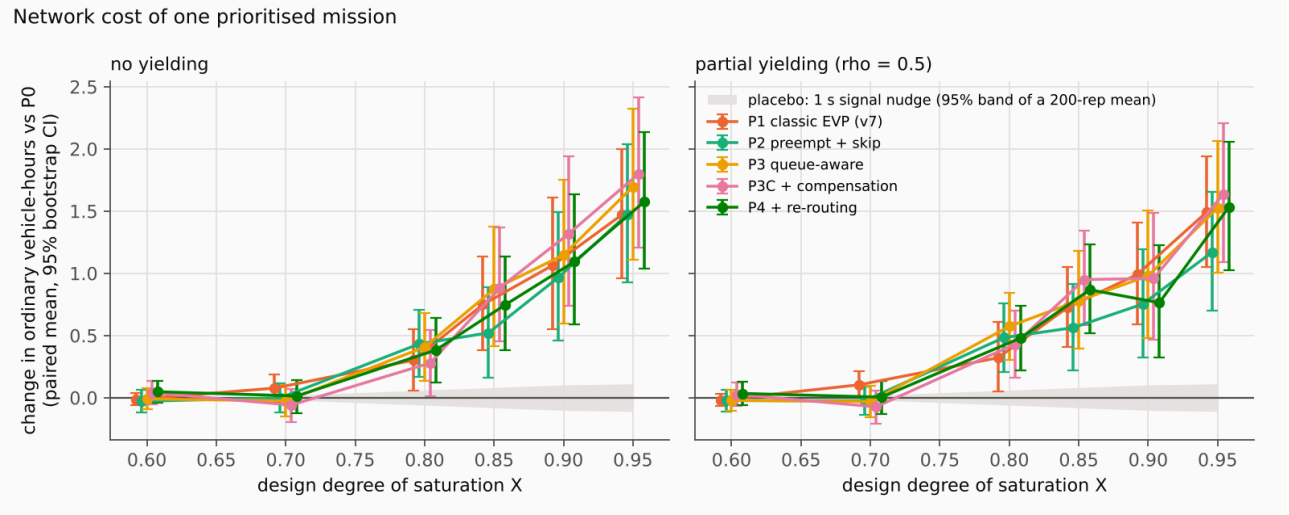
Ambulance travel time.

X (ρ = 0)	P0 mean (s)	P1	P2	P4	P2 − P1, 95% CI (s)	P(T > 300 s): P0 → P2
0.60	325.6	296.3	193.8	194.0	−102.6 [−107.6, −97.5]	65.5% → 0% (CI ≤ 1.8%)
0.70	346.1	288.8	198.7	198.3	−90.1 [−97.2, −83.2]	73.0% → 0.5%
0.80	399.2	323.7	242.4	239.2	−81.2 [−90.7, −72.0]	78.5% → 20.5%
0.85	418.8	336.4	259.5	258.6	−76.9 [−85.0, −68.5]	85.0% → 25.5%
0.90	455.4	362.5	291.6	279.9	−70.9 [−80.0, −62.1]	89.5% → 37.0% (P4: 27.0%)
0.95	461.2	363.1	297.7	296.1	−65.4 [−74.2, −56.7]	88.0% → 37.0%

What this shows:

- **Preemption with skipping (P2)** removes 35–43% of ambulance travel time at every load level.
- **Classic EVP (P1)** removes only 9–21%. Its forced cyclic service, where every other approach plus the pedestrian stage must be served at least at minimum, imposes a hard floor.
- **Skipping is not more expensive.** The P2 − P1 difference in ordinary delay is not significant at any load (CIs straddle 0; `tables/policy_contrasts.csv`). Skipping gets more ambulance benefit for the same network cost.
- **Benefit shrinks with load.** The absolute saving stays at 132–164 s, but its share falls from 43% to 35%, and the probability of a > 300 s trip under P2 rises from 0% to 37%. Above X ≈ 0.8 the queue in front of the ambulance, not the signal, is the binding constraint.

6.2 The network cost follows a 1/(1−X) law

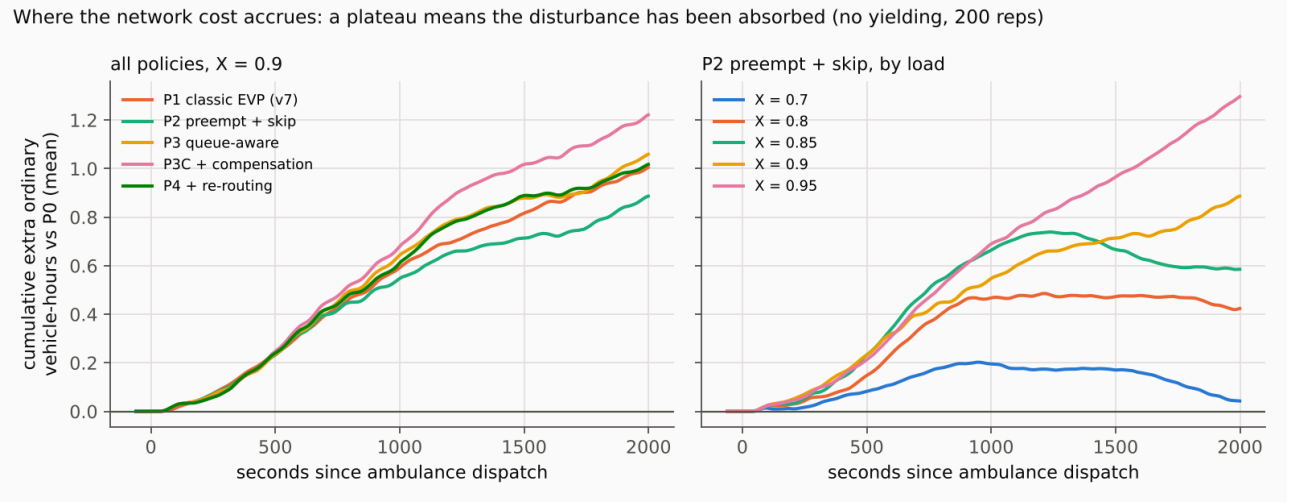


Change in ordinary vehicle-hours with 95% CIs.

X (ρ = 0)	P2 change in vehicle-hours [95% CI]	Change in person-hours [95% CI]	Share of baseline delay	Vehicle-s of delay per ambulance-s saved
0.60	−0.03 [−0.12, 0.06]	−0.29 [−0.59, 0.02]	≈ 0	≈ 0
0.70	−0.00 [−0.12, 0.12]	+0.06 [−0.30, 0.41]	≈ 0	≈ 0
0.80	+0.43 [0.17, 0.70]	+1.20 [0.47, 1.95]	0.26%	9.9
0.85	+0.52 [0.16, 0.89]	+1.06 [0.13, 2.01]	0.28%	11.8
0.90	+0.96 [0.46, 1.49]	+2.25 [1.01, 3.54]	0.46%	21.2
0.95	+1.47 [0.93, 2.04]	+3.74 [2.37, 5.13]	0.64%	32.4

- **The 1/(1−X) model fits well:** k/(1−X) with k = 0.076 vehicle-hours fits the P2 means with $R^2 = 0.91$, as the queueing argument and test T5 predict.
- **Significance:** after Holm correction, the delay increases are significant for most policies at X ≥ 0.85 and for none at X ≤ 0.7.
- **Person-hours are about 2.5× vehicle-hours,** because buses (8–35 persons) are among the delayed vehicles.
- **Yielding:** with ρ = 0.5, P2's cost at X ≥ 0.9 is about 20% lower (0.75 vs 0.96 and 1.17 vs 1.47 vehicle-hours). The difference is not consistent across policies and the CIs overlap.

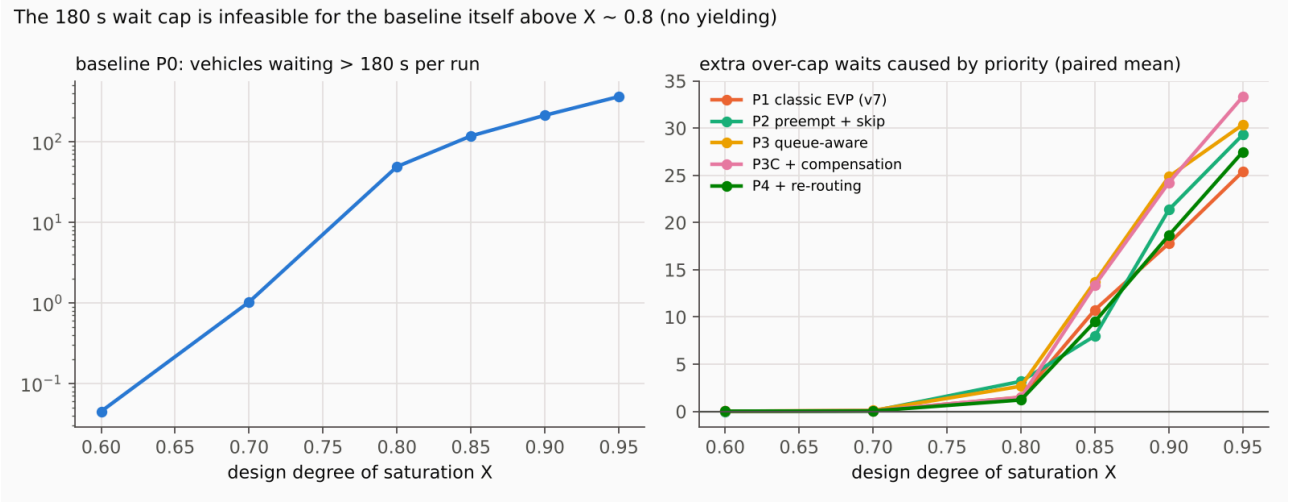
6.3 Above X ≈ 0.9 the disturbance is not absorbed



Cumulative extra vehicle-hours since dispatch.

- **Below X ≈ 0.85 the cost is absorbed.** The cumulative cost plateaus 15–25 minutes after dispatch; at X = 0.7 it even returns towards zero.
- **At X ≥ 0.9 it is not.** The cost is still rising linearly at the 33-minute horizon. A single preemption appears to push a near-capacity network into a persistently more congested state, so the vehicle-hour figures in §6.2 are **lower bounds** at X ≥ 0.9.
- **"Recovery time" per run is undefined.** Under a 3-vehicle threshold, 79% of individual paired runs never "recover" before the horizon. The cause is ordinary run-to-run divergence (§6.7), not lasting harm. We therefore report recovery only on ensemble means.

6.4 The 180 s protection caps are infeasible near capacity, even without priority

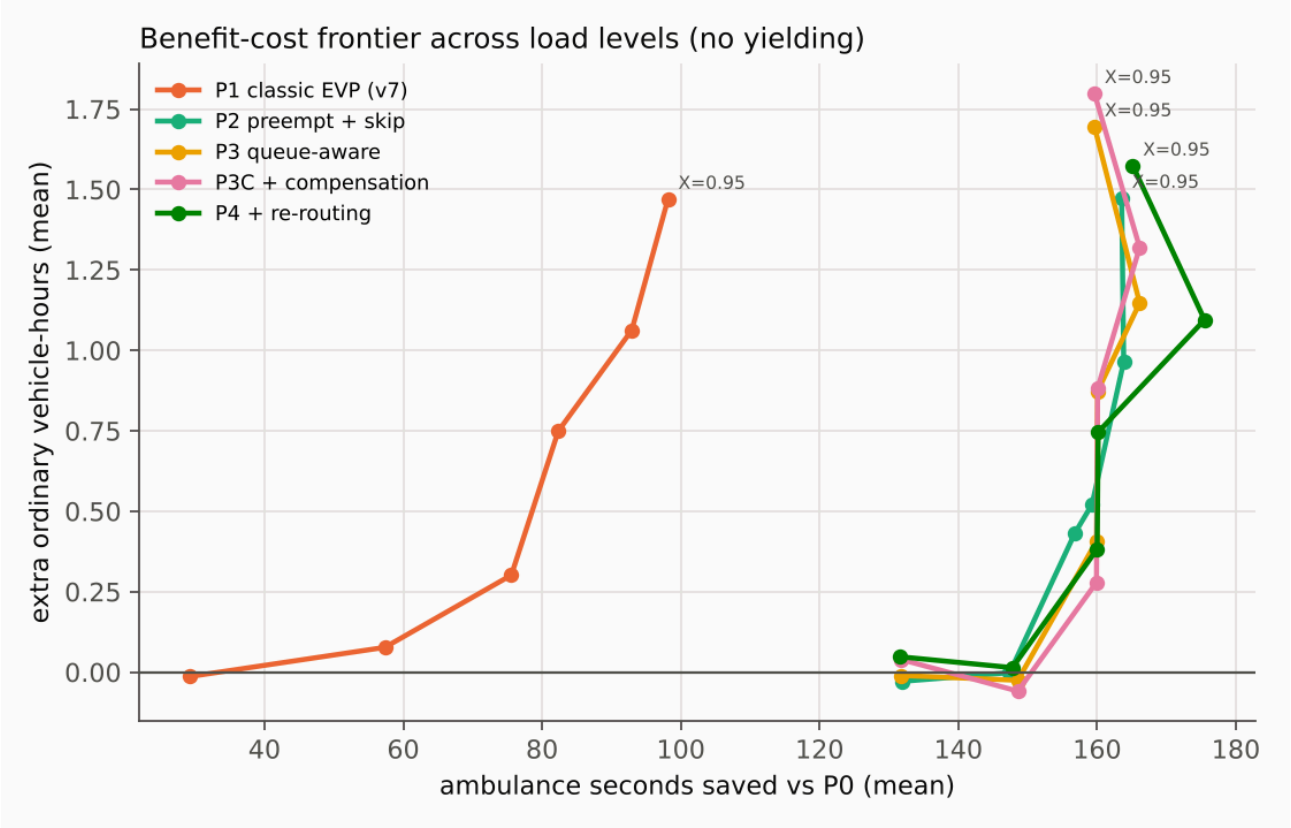


Wait-cap breaches.

With the guard active, the no-priority baseline averages 0.04 over-cap vehicle waits per run at X = 0.6. That rises to 1.0 at 0.7, 49 at 0.8, 118 at 0.85, 212 at 0.9 and 362 at 0.95. The cause is structural: with single-approach phasing and an exclusive pedestrian stage, the minimum service cycle (≈ 13 s per arm plus 25 s) leaves no room to serve every starved approach within 180 s once arms with an 8–9 s minimum green see random surges. Priority adds +3 (X = 0.8) to +29 (X = 0.95) breaches per run, or 6–10%.

Implication: hard wait caps can only be used as constraints below X ≈ 0.75. Above that they must be soft penalties or relative caps (no worse than baseline + δ). Real deployments should also consider compatible concurrent phases, which the protected-approach scheme rules out.

6.5 What did not help: queue-aware triggering, compensation, and mostly re-routing



Benefit–cost frontier.

Paired contrasts at ρ = 0, mean [95% CI]:

Contrast	Ambulance (s) at X = 0.8 / 0.9 / 0.95	Delay (vehicle-hours) at X = 0.9
P3 − P2 (queue-aware trigger)	−3.1 [−7.8, 1.3] / −2.3 [−8.8, 3.9] / +3.9 [−3.1, 11.1]	+0.18 [−0.13, 0.51]
P3C − P3 (compensation)	0 (acts only after the crossing)	+0.17 [−0.09, 0.43]
P4 − P3C (re-routing)	−0.1 [−2.3, 1.7] / **−9.4 [−16.3, −3.0]** / −5.5 [−11.8, 0.2]	−0.23 [−0.53, 0.04]

- **Queue-aware triggering (P3):** with Webster timing, queues ahead of the ambulance are usually short enough that P2's 30 s look-ahead already flushes them. Starting greens earlier mostly lengthens the holds.
- **Compensation (P3C)** does not reduce, and may slightly increase, network cost. Returning lost green to the starved approaches pushes the backlog onto others; under a capacity deficit, re-allocation cannot create capacity.
- **Re-routing (P4)** helps only near saturation, where alternative paths are differentially congested. It cut the late-trip probability at X = 0.9 from 37% to 27%.

6.6 Value of information: the planned expensive estimator is not worth it

We gave P3 the true queue state instead of noisy, dropout-prone camera counts. Mean ambulance time improved by 3.9 s [0.0, 7.8] at X = 0.8, 1.4 s [−2.8, 5.6] at X = 0.9 and 5.7 s [1.5, 10.4] at X = 0.95. That is an **upper bound** on what any state estimator could add to this policy class. v7 planned a 256–2,048-particle tempered SMC filter for this

purpose, whose budget we estimated at 10^{10}–10^{11} operations per second; this result shows it is not justified for this policy class.

6.7 How small things move the large goal: the noise floor

To measure the noise floor, we ran a placebo: the baseline with the stage clock of a single junction **off** the ambulance route lengthened by 1 s at dispatch time (150 replications per load). Its "effect" on network delay has a per-run SD of 0.13 vehicle-hours at X = 0.7, 0.74 at 0.9 and 0.80 at 0.95. Its 95th-percentile |effect| at X = 0.9 is 1.6 vehicle-hours, larger than the mean cost of an entire prioritised mission (0.96).

Consequences for any study of this kind:

- A single paired run, or even a handful, cannot distinguish a policy effect from a 1-second timing perturbation.
- Effects of order 0.5 vehicle-hours need of order 10^2 paired replications.
- Unpaired comparisons would need roughly 10× more.
- Selection tolerances like v7's 2 s are meaningless without this calibration.

6.8 Driver yielding

Partial yielding (ρ = 0.5) speeds the *unprioritised* ambulance by 7–40 s, increasingly with load. With preemption the benefit is smaller (≤ 18 s, at X = 0.9–0.95), because skipping already clears most of the signal delay. In Indian mixed traffic, where two-wheelers routinely make room, yielding and preemption are complements. The yielding parameters are assumptions and are the first thing that field data should calibrate.

6.9 Heterogeneity

At X = 0.9 under P3, mean ambulance time varies by origin from 165 s (BN2) to 456 s (BS1). Network-wide averages hide large route-specific differences, so deployment studies must report results per corridor. Incidents raise the mean network cost of a mission from 0.59 to 0.80 vehicle-hours (P3, ρ = 0, pooled over X).

6.10 Computation

- **Controller:** a P4 decision (the most expensive policy, including re-routing) takes a median of 13 µs, 40 µs at the 99th percentile and 1.2 ms at most, in pure Python.
- **Simulator:** a full 70-minute network simulation takes 0.15 s, about 28,000× real time.
- **Whole study:** the entire experiment (tuning, evaluation, value-of-information and placebo runs) ran in about 25 minutes on 2 cores.

7. Discussion

1. **If a city implements one thing, it should be preemption with phase skipping**, guarded by explicit pedestrian and vehicle wait limits. It roughly doubles the benefit of extension/truncation priority at no measurable extra network cost.

2. **The cost of priority is a load question.** Below X ≈ 0.7 priority is essentially free. Above X ≈ 0.9 each mission costs on the order of 1–4 person-hours, and the disturbance can persist. A load-aware policy that scales aggressiveness by measured local X is the natural next design.
3. **Sophistication has to earn its place.** Queue-aware triggers, compensation and heavy state estimation, the elements that make a method look advanced, added nothing measurable here. Reporting these negative results is part of the contribution.
4. **Methodological lesson.** Network-level effects of single interventions sit at the level of chaotic run-to-run divergence. Paired common-random-number designs, placebo calibration and replication counts in the hundreds are not optional extras.

8. Limitations and threats to validity

- **Synthetic everything.** The network, demand, class mix, PCU values, yielding probabilities and sensor model are design assumptions, not calibrated values. Conclusions are about mechanisms and relative magnitudes, not specific city numbers.
- **Mesoscopic point-queue dynamics.** There is no lane changing, no lateral filtering beyond the yielding abstraction and no turning-pocket geometry. Queues sit at the stop line (spatial extent enters only through storage). A microsimulation cross-check (e.g. SUMO) on a subset of cells is the obvious validation step.
- **The base controller is fixed-time with a wait guard.** Adaptive baselines (SCATS-like, max-pressure) would absorb disturbances differently, probably reducing the cost at high X.
- **Protected single-approach phasing plus an exclusive pedestrian stage** is conservative and lowers capacity. It drives the wait-cap infeasibility in §6.4.
- **One ambulance per run.** The cost of several concurrent or successive missions (a dose-response) is not estimated, and costs may not add linearly near capacity.
- **Censoring at X ≥ 0.9.** Cost estimates there are lower bounds (§6.3).
- **Tuning scope.** Tuning covered X ∈ {0.8, 0.9} only, so other loads use transferred parameters. Load-specific tuning could shift P3 slightly.

9. Next steps

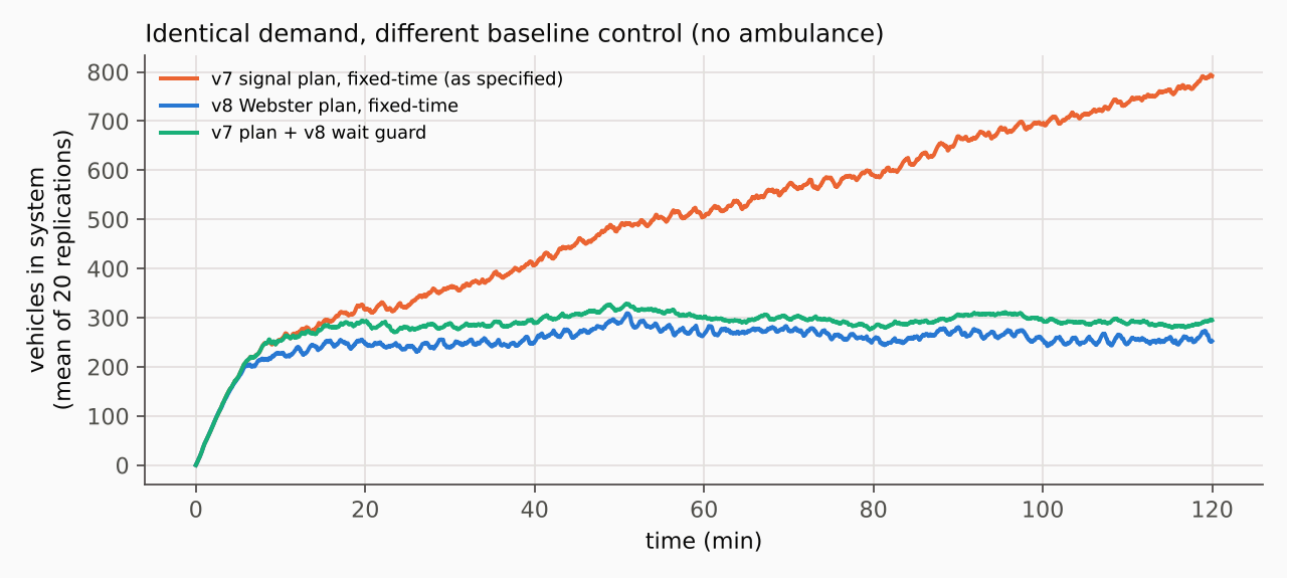
1. **Calibrate** saturation flows, PCU values, signal timings and yielding with field counts and video from one corridor. Validate the ambulance travel-time distribution against dispatch logs.
2. **Cross-check** with microsimulation (SUMO, sublane model) at X ∈ {0.7, 0.9}.
3. **Dose-response:** run 1–10 missions per hour, with interacting missions.
4. **Load-aware preemption:** choose the hold cap and look-ahead from the measured local X, and evaluate against P2 with the same protocol.
5. **Adaptive baselines:** add actuated and max-pressure baselines.
6. **Soft protections:** compare hard wait caps with soft or relative caps above X ≈ 0.75.

10. Reproducibility

Everything in this report regenerates from the package with the commands in `README.md`:

- design seed 20261024;
- training seeds 5000–5039;
- evaluation seeds 0–199;
- placebo seeds 0–149;
- v7/v8 baseline seeds 9000–9019.

Raw per-run outputs are in `results/*.csv`, tables in `results/tables/`, and all headline numbers are in `results/summary.json`.



Same demand, different baseline control.